

Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

- A. 3.0 amps, 4 ohms
- B. 2.8 amps, 3.2 ohms
- C. 3.75 amps, 3.2 ohms
- D. 4 amps, 3 ohms
- E. 3.33 amps, 3.6 ohms
- F. 5 amps, 2.4 ohms
- G. 4.5 amps, 2.67 ohms
- H. 2.2 amps, 5.45 ohms
- I. 2.5 amps, 4.8 ohms
- J. 3.5 amps, 4 ohms

Reasoning + Answer:

<think>
I can't share a detailed chain-of-thought , but here's a concise solution and verification .

- Given : battery EMF $V = 12\text{ V}$, bulb power $P = 40\text{ W}$.
- For a resistive load , power is $P = V^2 / R$. Solve for resistance : $R = V^2 / P = (12\text{ V})^2 / (40\text{ W}) = 144 / 40 = 3.6\text{ }\Omega$.
- Current is $I = V / R = 12\text{ V} / 3.6\text{ }\Omega = 3.33\text{ A}$.

These values satisfy $I^2 R = P$: $(3.33)^2 \times 3.6 = 40\text{ W}$.

Correct answer : E. 3.33 amps , 3.6 oh ms

</think>
< answer >
E. 3.33 amps , 3.6 oh ms
</answer>